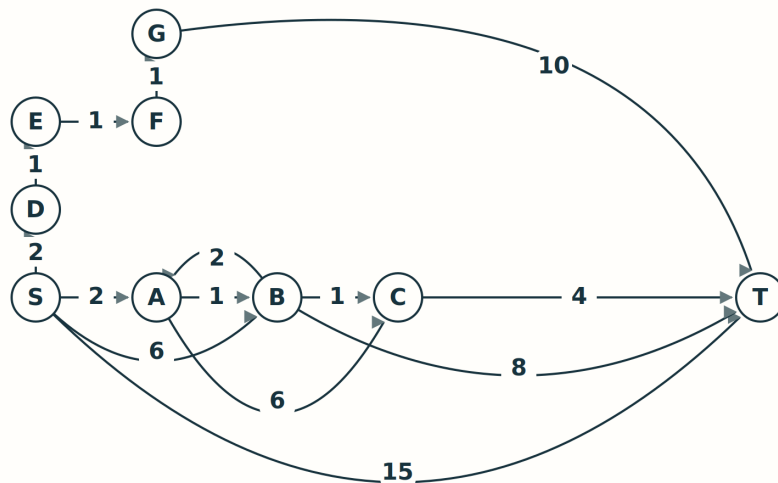


# Dijkstra and A\*: worksheet

Name: \_\_\_\_\_ Partner: \_\_\_\_\_

## 1. Which routes must we keep? | 09-28 min

Directed edges carry additive costs. Start at S and finish at T. Use the same graph throughout this page. Do not assume a discovered distance is final.



**Predict:** What does first discovery of T return? Find a cheaper candidate route.

First discovered cost: \_\_\_\_\_ Cheaper route and cost: \_\_\_\_\_

**Trace the first six non-stale extractions.** Write only labels that improve. Tie-breaking: neighbor order from the fixture; FIFO for equal priorities. The first two non-stale extractions are S, then A.

| Selected node and g | Improved labels and their predecessors |
|---------------------|----------------------------------------|
| S, 0                |                                        |
| A, 2                |                                        |
|                     |                                        |
|                     |                                        |
|                     |                                        |
|                     |                                        |
|                     |                                        |

**Justify:** Why can the minimum-g frontier node be finalized? Where is nonnegativity used?

# Counterexample laboratory

## 2. Is admissibility enough to close a state forever? | 61-68 min

Use this separate four-node graph; all edges are directed. The heuristic values are assigned estimates, not drawing distances.

$S \rightarrow A : 3$        $S \rightarrow B : 1$        $B \rightarrow A : 1$        $A \rightarrow T : 3$   
 $h(S) = 0$        $h(A) = 0$        $h(B) = 4$        $h(T) = 0$

Confirm that every  $h$  value is admissible. Then trace  $A^*$  using  $f=g+h$ .

| Selected state | $g$ | $h$ | $f$ | What changes? |
|----------------|-----|-----|-----|---------------|
| S              |     |     |     |               |
|                |     |     |     |               |
|                |     |     |     |               |
|                |     |     |     |               |
|                |     |     |     |               |

Returned cost without reopening: \_\_\_\_\_ With reopening: \_\_\_\_\_

Which edge violates consistency? Write the failed inequality.

Repair the implementation, or strengthen the assumptions. State both options.

## 3. Change the movement rules | 77-86 min

On a four-neighbor unit grid, removing walls motivates Manhattan distance. Now allow a diagonal move for cost 1. Give a start/goal pair that refutes Manhattan admissibility.

What lower bound would fit the new movement model?

## Exit ticket | 86-90 min

**A.** Why is discovering the goal different from certifying its cost?

**B.** Why is skipping a stale entry different from refusing to reopen a state?

**C.** Does a smaller number of expansions prove a smaller runtime? Explain one missing factor.